EMMANUEL WARUTERE WACHIRA

Full-Stack & AI Engineer | DevOps Engineer

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Portfolio: wew-portfolio.vercel.app • GitHub: github.com/wachira7 • LinkedIn: linkedin.com/in/emmanuelwaruts77

EXECUTIVE SUMMARY

Full-Stack and AI Engineer who builds and ships production systems end to end. Recently built The Pour Is Right, an LLM application that prices wine with a QLoRA fine-tuned open-source model, RAG and an ensemble, served on Modal, with LangGraph agents that find live deals and send Telegram alerts. Also built and deployed LogiFlow, a multi-tenant logistics SaaS, and FlixVideo, an AI-powered streaming platform with Claude recommendations, both with CI/CD, monitoring and Stripe, M-Pesa and crypto payments. Backed by 5+ years of field engineering at KOKO Networks during its growth from 150,000 to 1.5 million households. Certified in AI engineering (Udemy, Ed Donner), AWS Cloud Architecting, Google Project Management and Six Sigma Green Belt.

TECHNICAL SKILLS

AI & LLM Engineering: OpenAI, Anthropic Claude, Groq and Hugging Face APIs; LangChain; RAG with Qdrant and Chroma; QLoRA fine-tuning (Transformers, PEFT, TRL, bitsandbytes); agents with LangGraph, tool calling and structured outputs; GPU model serving on Modal; PyTorch, scikit-learn, XGBoost; Gradio
Languages: Python, JavaScript, TypeScript, Solidity, Bash
Frontend: React.js, Next.js, Tailwind CSS
Backend: Django REST Framework, Node.js, Express.js, Celery, BullMQ, FastAPI
Databases: PostgreSQL, MongoDB, Redis, SQLite
DevOps & Cloud: Docker, GitHub Actions, AWS (S3, EC2, RDS, IAM), Oracle Cloud, Kubernetes, Terraform, Prometheus, Grafana
Payments: Stripe, M-Pesa (Daraja API), NOWPayments (Crypto)
Storage: AWS S3, Cloudinary
Tools: Git, GitHub, Sentry, Yarn, Swagger, Leaflet.js, Socket.io
PM Tools: Asana, Notion, Jira, Agile/Scrum, Six Sigma

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer | DevOps Engineer — *WaruTech | 2025 — Present*

- Architected and built LogiFlow, a multi-tenant logistics SaaS platform using Django REST Framework and Next.js, featuring 8 modules: inventory, warehousing, production, distribution, billing, analytics, company management, and platform administration.
- Engineered FlixVideo, an AI-powered movie and TV discovery platform using Node.js (Express) and Next.js, integrating Claude AI recommendations, real-time watch parties via Socket.io, and a complete subscription system across three payment gateways.
- Implemented full CI/CD pipelines using GitHub Actions for both platforms; deployed LogiFlow to Northflank and FlixVideo to Render/Vercel, incorporating automated migrations and Docker containerization for zero-downtime deployments.
- Deployed production monitoring stacks (Prometheus + Grafana) on Oracle Cloud Free Tier VM, enabling real-time visibility into application health, API response times, Redis metrics, and BullMQ job queues.

Field Service Engineer — *KOKO Networks | Feb 2020 — Jan 2026*

- Contributed to scaling KOKO Networks' customer base from 150,000 to 1.5 million households across Kenya over 5 years — a 10x growth trajectory — by maintaining 90%+ uptime across an assigned zone of 15 KOKO Points (out of 3,000 nationwide), ensuring consistent clean fuel availability across Nairobi and the Lower Mount Kenya region.
- Led a team of 6 technicians at the KOKO repair centre for one year, overseeing burner production operations to maintain 100% parts availability for the cooker assembly line and refurbishment, contributing to approximately 11,000 cooker units serviced against a 25,000-unit annual target under operational constraints.
- Drove product durability improvements by providing post-development engineering insights on material changes for KOKO Point components, resulting in a documented reduction in wear-and-tear and a durability success rate exceeding 80%.

- Executed systematic preventive maintenance, quality assurance checks, and technical troubleshooting across smart fuel dispensing infrastructure, applying structured problem-solving methodologies aligned with Six Sigma quality standards.

KEY PROJECTS

The Pour Is Right — LLM Wine Pricer and Deal-Hunting Agents | *Python + PyTorch + Hugging Face + Qdrant + LangGraph + Modal + Gradio 2026*

- Curated 280,000 wine reviews into a 22,000-item dataset and rewrote it with an LLM through the Groq Batch API for about $1, cutting price leakage in the text by 93%.
- Fine-tuned Qwen 3 4B with QLoRA on a free Colab T4, reducing its error from $23.15 to $10.10, level with Claude Sonnet 5; diagnosed a mixed-precision issue that made training six times faster.
- Built RAG on a Qdrant vector store and an ensemble of Claude, the fine-tuned model and a PyTorch network: average error $7.96, 63% below the baseline.
- Deployed the model as a GPU service on Modal, and built agents in LangGraph (fan-out, conditional routing, human-in-the-loop approval) that scan live wine deals and send Telegram alerts, with a Gradio front end.

Code available on request

LogiFlow — Multi-Tenant Logistics SaaS | *Django + Next.js + PostgreSQL + Docker 2025 — Present*

- Production-grade multi-tenant platform with role-based access, automated billing (Celery) with Stripe/M-Pesa/crypto payments, and real-time distribution tracking via Leaflet.js and OpenStreetMap/Nominatim geocoding — including lat/lng coordinate storage and geospatial route planning on PostgreSQL, directly analogous to PostGIS spatial workflows.
- Implemented centralized auth guard architecture, JWT token management, and a platform admin module for tenant management, subscription oversight, system health, and audit logging.

Live: logi-flow-three.vercel.app GitHub: github.com/wachira7/LogiFlow

FlixVideo — AI-Powered Streaming Platform | *Node.js + Next.js + PostgreSQL + Redis + BullMQ + MongoDB 2025 — Present*

- Full-featured streaming and discovery platform with Claude AI recommendations, Redis-cached search, watch party synchronization, subscription tiers, and 100+ REST API endpoints. Integrated MongoDB (native driver) for four real-time data domains: watch party chat messages, user watch history, AI conversation persistence, and in-app notifications, using a shared getCollection helper pattern for lightweight, schema-free document storage.
- Provisioned AWS S3 and IAM resources using Terraform (remote state managed in S3 + DynamoDB); resolved production Redis connection limits by refactoring BullMQ to shared IORedis connections; migrated from Upstash to Redis Cloud; deployed Prometheus + Grafana monitoring on Oracle Cloud VM.

Live: flix-video-ten.vercel.app GitHub: github.com/wachira7/Flix-video

Vehicle Shield Insurance DApp | *Solidity + Web3.js + Next.js + Hardhat + Firebase 2025*

- Blockchain-based vehicle insurance platform with Solidity smart contracts on Ethereum, Web3 wallet integration, automated claims processing, and Firebase Auth for user management.

Live: vsi-app.vercel.app GitHub: github.com/wachira7/vehicle-shield-insurance

EDUCATION

BSc Mechatronic Engineering — Second Class Honours

Jomo Kenyatta University of Agriculture & Technology (JKUAT) | 2013 — 2018

Software Engineering Certification — Full-Stack Developer

Moringa School | Feb — Nov 2024

CERTIFICATIONS

- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents — Udemy, Ed Donner (Sep 2026) — ude.my/UC-d4dbf930-d272-49e9-bdfc-9ae7609320ed
- AWS Academy Graduate — Cloud Architecting — Amazon Web Services (Jul 2025)
- AWS Academy Graduate — Cloud Foundations — Amazon Web Services (Jun 2025)
- DevOps Engineering Certification — Nairobi DevOps Community (Nov 2025)
- Google Project Management Certificate — Coursera / Google (Aug 2022)
- Six Sigma Green Belt — International Six Sigma Institute (Jul 2020)

References available on request.